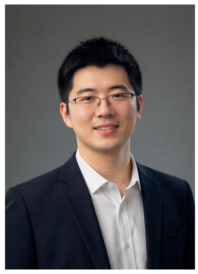

YINDA CHEN

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https://ydchen0806.github.io/

https://scholar.google.com/citations?user=hCvlj5cAAAAJ&hl=en&oi=ao

EDUCATION

University of Science and Technology of China & Shanghai AI Lab **Hefei & Shanghai**

Information and Communication Engineering, Ph.D. Candidate **2024.7 ~ 2027.6 (Expected)**

- **Research Focus:**Predictive multimodal intelligence: from representation and predictive learning to world models for embodied perception, future prediction, and action generalization
- **Advisors:**Prof. Feng Wu (CAE Academician, IEEE Fellow), Prof. Zhiwei Xiong; Co-supervisor: Prof. Xiaoou Tang (IEEE Fellow, Shanghai AI Lab)
- **Core Courses:**Algorithm Design and Analysis, Statistical Learning, Deep Learning, Reinforcement Learning
- **Honors:**Principal Investigator of National Natural Science Foundation Ph.D. Project (2024)

University of Science and Technology of China **Hefei**

Computer Technology, M.S. **2022.9 ~ 2024.7**

- **Honors:**National Graduate Scholarship (2022)

Xiamen University **Xiamen**

Environmental Ecological Engineering & Economics (Dual Degree), B.S. **2018.9 ~ 2022.7**

- **Overall Ranking: 1/31**
- **Honors:**Xiamen University Academic Star (2021), CDA Level 1 Certification (2022), Kaggle Expert
- **Advisor:**Prof. Yuanye Zhang

RESEARCH PROJECTS

Embodied Intelligence & World Models: Pelican Series **2026.5 ~ Present**

Unified Embodied Foundation Models *Core Contributor (Ranked Second); VLA Technical Report*

- In **Pelican-Unified 1.0**, helped build a unified embodied model for **understanding, reasoning, imagination, and action**, integrating multimodal understanding, future video rollout, and low-level action generation.
- As the concrete **VLA component** within the Pelican-Unified framework, contributed to the **Pelican-VLA 0.5: Attending Before Acting Benefits Generalization** technical report, unifying vision-language understanding, **future-frame generation**, and action prediction; learnable **Bottleneck Tokens** support cross-scene and cross-embodiment generalization.
- Together, these technical reports connect **unified multimodal representations**, **future-state modeling**, and **closed-loop action** into a world-model narrative. Pelican-Unified 1.0; Pelican-VLA 0.5

National Natural Science Foundation Ph.D. Project **2025.1 ~ 2027.12**

NSF Project *Principal Investigator (Funding: 300k RMB, Only recipient in Information field, Anhui Province, 2024)*

[1] Foundation **10-billion-scale** brain neural **foundation model** training
- Neuron underlying visual feature **token construction**
- Neural architecture search, **autoregressive visual model** training

[2] **Fine-grained** neuron **segmentation**
- **Skeleton features**, attention decomposition mechanisms
- Complementary masking, segmentation **robustness enhancement**

[3] Neural **circuit reconstruction**
- Point cloud morphological features and image feature **alignment**
- Candidate competition strategy, **progressive reasoning**, long-range tracking

Representation and Predictive Learning **2022.5 ~ 2023.12**

Conference&Journal *First Author*

[1] **TokenUnify: Scalable Autoregressive Visual Pre-training with Mixture Token Prediction**, **ICCV 25** **[CCF-A]**
- Proposed image autoregressive pretraining combined with Mamba framework, demonstrating advantages of long sequences and low computational cost.
- Demonstrated good scaling laws with corresponding theoretical proof, showing logarithmic-linear relationship between model performance and parameters.

[2] **MaskTwins: Dual-form Complementary Masking for Domain-Adaptive Image Segmentation**, **ICML 25** **[CCF-A]**
- Proposed new complementary masking theory from sparse signal reconstruction perspective, rigorously proving theoretical advantages of dual-form complementary masking in extracting domain-invariant features.
- Designed simple and efficient unsupervised domain adaptation framework, achieving cross-domain image segmentation through complementary masking consistency learning, improving natural image segmentation by 2.7% mIoU and biomedical image segmentation by 3.2% IoU.

[3] **【LONG ORAL】** **Self-supervised neuron segmentation with multi-agent reinforcement learning**, **IJCAI 23** **[CCF-A]**

- Improved MAE masking strategy using reinforcement learning, automatically selecting mask ratios and schemes.
- Introduced multi-agent framework for more efficient self-supervised feature learning, improving segmentation accuracy by 12%.

[4] Learning multiscale consistency for self-supervised electron microscopy instance segmentation, **ICASSP 24** **[CCF-B]**
- Achieved high-performance pretraining strategy based on multiscale feature contrastive learning and feature reconstruction.
- Proposed feature consistency loss function, overcoming representation differences caused by scale variations, improving accuracy by 9%.

Multimodal and Generative Models — 2023.8 ~ Present

Conference&Journal *First Author & Co-first Author*

[1] 【Accepted】Learned Image Coding with Generative Reference of Conditional Latents, **TPAMI** **[SCI-Q1]**
- Extension work of AAAI 25 oral paper, further exploring benefits of reference images for image coding.
- Obtained reference images through local dictionary, network retrieval, and image generation, improving compression performance by 15%.
- Proposed conditional latent generation framework, effectively solving performance degradation when reference images are unavailable.

[2] 【ORAL】Conditional Latent Coding with Learnable Synthesized Reference for Deep Image Compression, **AAAI 25** **[CCF-A]**
- Built image similarity dictionary for retrieving similar images to improve entropy model probability estimation.
- Proposed learnable synthetic reference framework, improving PSNR by 0.6dB at the same bit rate with leading compression performance.

[3] [Major Revision] Generative Text-Guided 3D Vision-Language Pretraining for Unified Medical Image Segmentation, Submit to PR
- Generated image descriptions using large language models for multimodal image-text contrastive learning pretraining.
- Introduced generative text guidance to address scarce medical-image annotations and enable zero-shot segmentation.

[4] BIMCV-R: A Landmark Dataset for 3D CT Text-Image Retrieval, **MICCAI 24** **[CCF-B]**
- Built the first open-source 3D CT image-text pair dataset containing 10,000 high-quality medical image-description pairs.
- Achieved efficient image-text information retrieval and keyword search, improving recall rate by 25% over traditional methods.

[5] MaskFactory: Towards High-quality Synthetic Data Generation For Dichotomous Image Segmentation, **NeurIPS 24** **[CCF-A]**
- Generated corresponding mask-image pairs using ControlNet through rigid and non-rigid deformation editing of masks.
- Core techniques: deformation-driven mask editing, ControlNet-guided image generation, synthetic data quality assessment.
- Synthetic data performance approached real data in downstream segmentation tasks with only 2% performance gap, significantly reducing annotation costs.

[6] [Major Revision] Joint Semantic and Coded Generation for Conditional Latent Coding, Submit to TCSVT
- Proposed joint semantic and coded generation framework, combining text descriptions with minimal coded features (0.01 bpp) to guide controllable rectified flow models in generating high-fidelity reference images.
- Core techniques: ControlNet for DiT architecture, iterative reference alignment (IRA), dynamic feature fusion (DFS), iterative probability refinement (IPR), theoretical robustness proof.
- Improved compression performance by 15.8% BD-rate compared to VVC, successfully unifying image generation and compression, bridging the gap between semantic and coded representations.

[7] EMPOWER: Evolutionary Medical Prompt Optimization With Reinforcement Learning, **JBHI** **[SCI-Q1]**
- Proposed the first evolutionary prompt optimization framework for medical domain, combining domain knowledge and reinforcement learning.
- Core techniques: medical attention, multi-dimensional quality assessment, evolutionary search, and semantic verification for clinical guideline consistency.
- Reduced factual errors by 24.7%, improved domain specificity by 19.6%, and achieved 15.3% higher clinician preference in blind testing.

Large Model Engineering — 2023.9 ~ Present

Projects *Core Member*

- **Foundation-model systems:** led a billion-parameter image-coding architecture with 30% higher compression than traditional standards; built LLM-driven quantitative-research workflows with 10k-GPU-scale training experience.
- **Embodied and biomedical AI:** contributed to real-robot closed loops, Unify pretraining, and embodied-AI infrastructure on 100-GPU H200 clusters; led medical-image and neuron-model pretraining with 64 A40 GPUs, DDP, and DeepSpeed for 30B-parameter-scale optimization.

Industry Internship Experience

Ubiquant — Beijing — 2026.5 ~ Present

Quantitative Research and Large Language Models *Researcher*

- Working on **quantitative research** and **general large models**, targeting **DeepSeek**-level capabilities.
- Experienced with **10k-GPU-scale clusters** and exploring LLM applications in factor mining and data-driven decision making.

Beijing Humanoid Robot Innovation Center (UBTECH) Beijing **2025.12 ~ Present**

Embodied Intelligence World Model Algorithm Team *Core Member*

- Developed **unified embodied world models** for humanoid robots, connecting task-relevant perception, future-state prediction, and low-level action generation in real-robot closed loops.
- Contributed to **Unify** pretraining and planning validation; investigated its VLA bottleneck representations for cross-scene and cross-embodiment manipulation generalization.
- Experienced with **100-GPU H200** cluster training for large-scale embodied foundation models.

Tencent (IEG) Shanghai **2025.8 ~ 2025.12**

Qingyun Program *Research Intern*

- Worked on **game world models** for game scene understanding, behavior modeling, and content generation.
- Delivered the **Honor of Kings Lingbao** project, focusing on **TTS** and **real-time match commentary**.
- Trained and optimized Lingbao speech generation models based on f5-tts and indextts2.

PLA General Hospital (301 Hospital) Beijing **2023.9 ~ 2024.2**

Data Compression Group *Research Intern*

- Collaborated with Academician Qionghai Dai's team on efficient data compression research.
- Designed medical imaging-specific compression algorithms optimized for CT, MRI modalities, improving compression efficiency by 35%.

Academic Internship Experience

Imperial College London London (Remote) **2022.11 ~ 2023.8**

Data Science Institute *RA*

- Collaborated with Associate Professor Rossella Arcucci on multimodal pretraining research, submitting one journal paper.
- Developed image-text contrastive learning framework, achieving 93.5% accuracy on medical diagnosis tasks.

Xiamen University WISER Club Xiamen **2021.8 ~ 2022.7**

Data Mining Group *Academic Community Member*

- Responsible for designing and discussing data mining courses, teaching clustering and Transformer sections.
- Mentored 20 undergraduate students to complete machine learning projects, organized 2 campus competition activities.

Wang Yanan Institute of Economics, Xiamen University Xiamen **2020.8 ~ 2021.12**

Econometrics *Research Assistant*

- Assisted Associate Professor Jiong Zhu in completing territorial economic statistics, conducting visual feature extraction of homestead information.
- Developed satellite image analysis tools for automatic land use change identification with 85% accuracy.

Honors and Awards

- **National Natural Science Foundation Ph.D. Project** **2024.12**
  - Principal Investigator, Only recipient in Information field, Anhui Province
- **Interdisciplinary Contest in Modeling (ICM), Outstanding Winner** **2024.05**
  - Top 0.17% of 10,388 participating teams worldwide
- **National Graduate Scholarship** **2022.12**
  - Award rate 1%
- **Xiamen University Academic Star** **2021.12**
  - Only undergraduate recipient
- **'Jingrun Cup' Mathematics Competition** **2021.09**
  - First Place, Xiamen University
- **'Internet+' Innovation Competition** **2021.08**
  - Gold Award, Fujian Province
- **National College Student Mathematics Competition Finals** **2021.05**
  - National Second Prize
- **'Challenge Cup' Academic Competition** **2021.05**
  - First Prize, Fujian Province
- **National College Student Mathematics Competition** **2020.11**
  - First Place, Fujian Province

Technical Skills

- **Programming**: Python, C, C++, Java, MATLAB, LaTeX
- **Deep Learning**: PyTorch, TensorFlow, DeepSpeed, DDP
- **Languages**: English (TOEFL 110, GRE 328), Chinese (Native)
- **Tools**: Git, Docker, CUDA, HPC

Service: Reviewer for CVPR'25, ICCV'25, WACV'26, NeurIPS'24, ICML'25, ICLR'24, MICCAI'25, ACM MM'24, AISTATS'24, IJCV, and TIP.